

Power BI Dashboard Report

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1. Introduction

This report presents the Power BI tasks developed as part of coursework, including both initial developments and subsequent updates. The aim is to analyze data using visualization techniques and generate meaningful insights to support decision-making.

2. Task 1: Dashboard Development (Original)

2.1. Objective

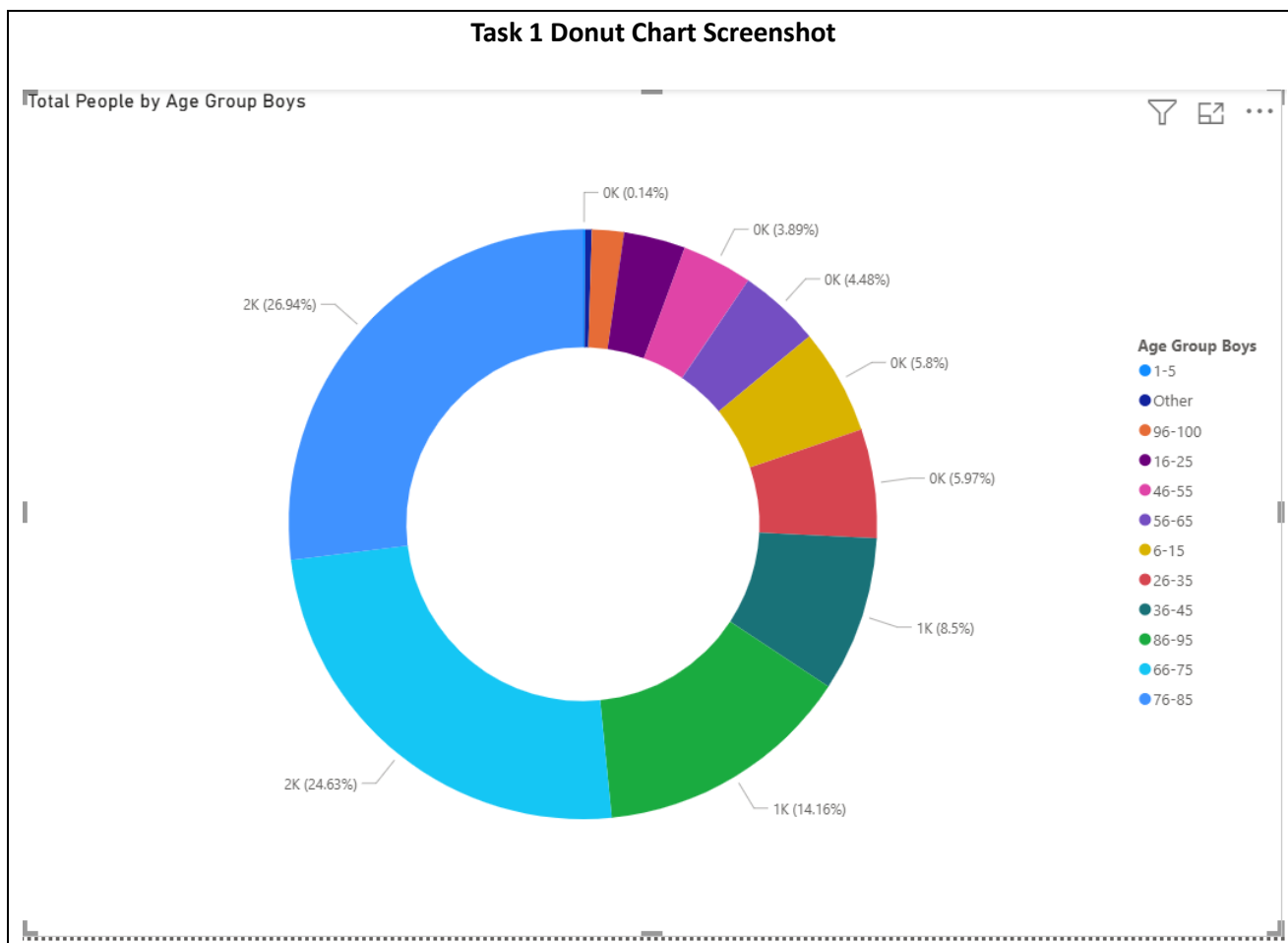
The objective of Task 1 was to explore the dataset, clean it, and create meaningful visualizations to represent demographic distributions.

2.2. Data Cleaning

Unnecessary columns were removed, missing values were handled, and data types were corrected to ensure accurate analysis of age groups and population metrics.

2.3. Visualizations

Different charts were created to represent patterns and trends in the dataset, focusing on the "Total People by Age Group Bays".



2.4. Insights

The visualizations identified key trends in age distribution, showing high density in the 16-35 and 40-55 age brackets.

3. Task 2: TAT Analysis Dashboard (Original)

3.1. Objective

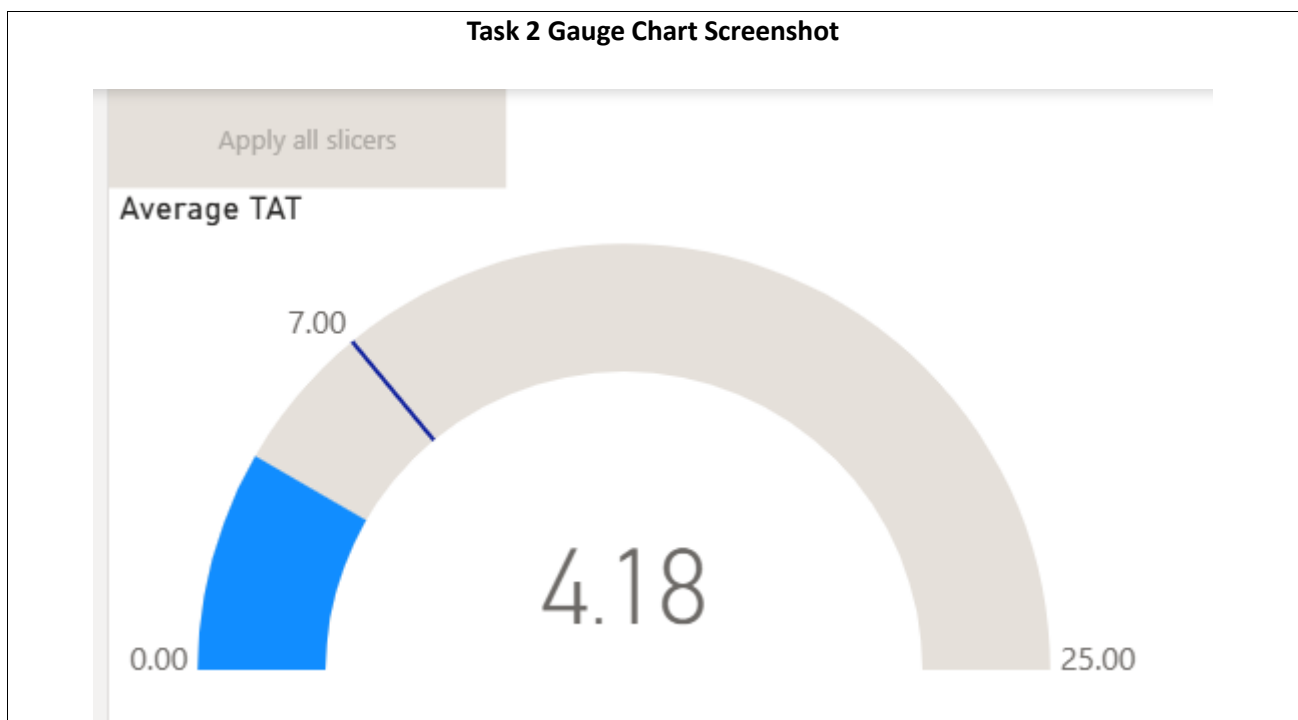
The objective of Task 2 was to calculate Turnaround Time (TAT) and analyze performance against a target benchmark of 7.00.

3.2. Data Processing

A new column 'TAT' was created by subtracting date fields. Measures such as sum, count, and average were calculated.

3.3. Visualizations

A gauge chart was created to represent the Average TAT (4.18) along with min (0.00), max (25.00), and target (7.00) values.



4. Updated Tasks: Dynamic Time & Real-Time Tracking

4.1 Objective

The objective of the update was to implement "Last Refreshed" timestamps using Direct Query (M Code) and dynamic time displays using DAX to ensure the dashboard reflects the most current data state.

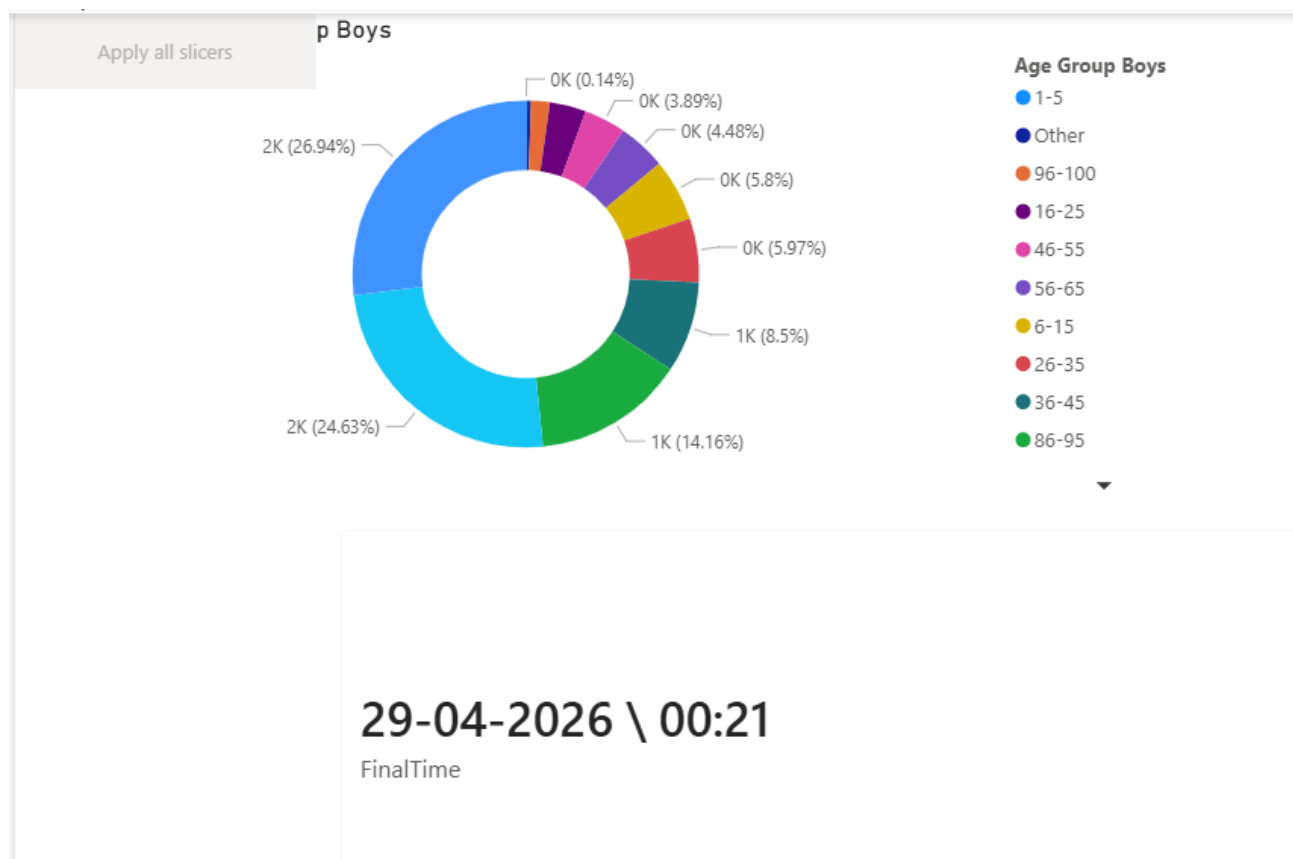
4.2 Task 1 Update: Dynamic Time Display

For Task 1, a simple dynamic time measure was implemented to show the current system time on the dashboard.

DAX Implementation:

```
CurrentDateTime = NOW()
```

Task 1 Updated Screenshot



4.3 Task 2 Update: Direct Query & Advanced DAX

Task 2 was enhanced with a "Last Refreshed" table using Power Query and a formatted DAX measure for the dashboard header.

Step 1: Power Query (M Code) for EST Refresh Time To capture the exact refresh time in EST, the following M Code was used in the Advanced Editor:

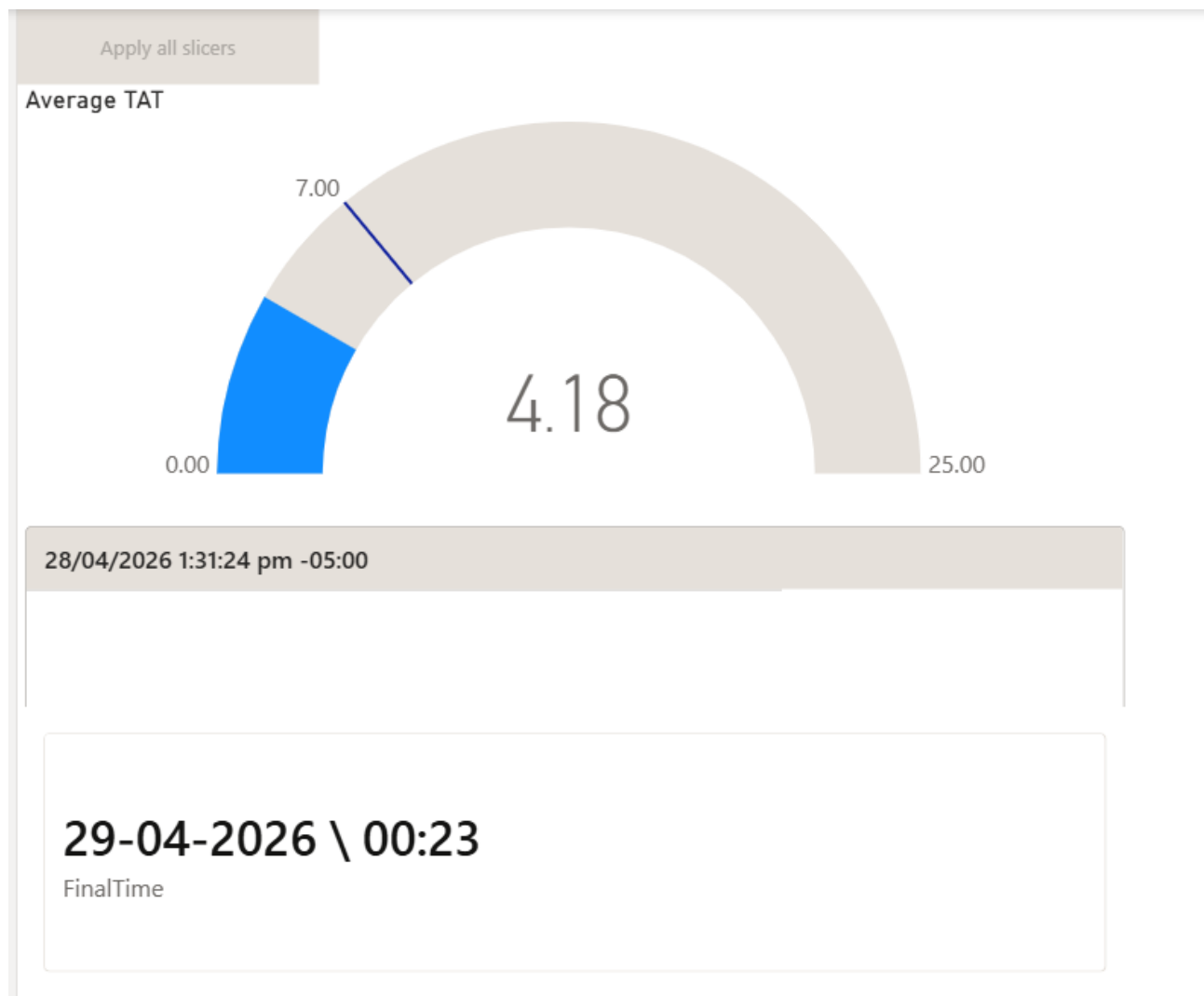
```
let
    Source = DateTimeZone.UtcNow(),
    EST_Time = DateTimeZone.SwitchZone(Source, -5),
    ConvertToTable = #table(
        {"Last Refreshed (EST)"},
        {{EST_Time}}
    )
in
    ConvertToTable
```

Step 2: Advanced DAX for Formatted Time A combined measure was created to split and format the Date and Time for the UI:

```
DatePart = DATEVALUE(NOW())
TimePart = TIMEVALUE(NOW())
HourPart = HOUR(NOW())
MinutePart = MINUTE(NOW())

FinalTime =
FORMAT([DatePart], "dd-MM-yyyy")
& " \ "
& FORMAT([HourPart], "00")
& ":"
& FORMAT([MinutePart], "00")
```

Task 2 Updated Screenshot (Gauge + Refreshed Time)



5. Conclusion

The updates to Task 1 and Task 2 demonstrate advanced Power BI capabilities. By integrating Direct Query timestamps and DAX time functions, the dashboards now provide not only historical insights but also real-time context regarding data freshness and system time, facilitating more accurate decision-making.